



MCAST

Hybrid Vision-Based Traffic Anomaly Detection Using Vehicle Tracking and Speed Estimation

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Technology in partial fulfilment of the requirements for the degree of BSc (Hons)
Multimedia in Software Development**

Authorship Statement

This dissertation is based on the results of research carried out by myself, is my own composition, and has not been previously presented for any other certified or uncertified qualification.

The research was carried out under the supervision of Marco Farrugia

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Abstract

The rapid expansion of urban infrastructure necessitates the implementation of robust Intelligent Transport Systems (ITS) to mitigate traffic accidents and congestion. However, traditional vision-based monitoring systems often struggle to balance precise spatiotemporal detection of complex collisions with the stringent latency requirements inherent to real-time edge computing. To address this challenge, this study proposes a novel, hybrid vision-based framework that structurally separates spatial vehicle tracking from kinetic anomaly classification. The architecture utilises a pre-trained YOLO11 model and BoT-SORT for persistent spatial tracking, combined with Inverse Perspective Mapping (IPM) and Kalman filtering to estimate three-dimensional metric speeds. Simultaneously, an asynchronous kinetic thread captures unstructured physical motion employing Farnebäck dense optical flow, which feeds spatiotemporal tensors into a custom-trained ResNet-18 Convolutional Neural Network. The system's performance, evaluated against the CADP and TAD benchmark datasets, demonstrated an impressive overall accuracy of 96.61

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List of Abbreviations

NN	Neural Network
ML	Machine Learning
DL	Deep Learning
FCN	Fully Convolutional Network
CNN	Convolutional Neural Network

Chapter 1: Introduction

1.1 Background and Context

The rapid and ongoing expansion of global urban infrastructure has led to an unprecedented rise in vehicular traffic. As a result, this increased density has heightened the frequency of road traffic collisions and systemic congestion, thereby posing substantial risks to public safety and economic efficiency. In response to these escalating challenges, contemporary municipal authorities are progressively adopting Intelligent Transport Systems (ITS) to protect public infrastructure and optimise traffic management. Traditionally, traffic monitoring has depended heavily on intrusive hardware, such as inductive loop detectors, or manual human observation. However, these conventional methods present notable operational limitations; physical sensors are unable to furnish comprehensive spatial context, while human observation remains inherently vulnerable to fatigue-induced errors and delayed response times. To address these shortcomings, the integration of deep learning and computer vision technologies into existing Closed-Circuit Television (CCTV) networks has emerged as a highly effective, non-intrusive approach for collecting high-dimensional data concerning vehicle trajectories, interactions, and anomalies.

1.2 Problem Statement

Notwithstanding these advancements, the task of robust traffic anomaly detection continues to constitute a highly intricate challenge within the field of computer

vision. Current object detection architectures, although highly effective at isolating individual vehicles under ideal conditions, often prove inadequate when faced with the chaotic and unstructured spatial dynamics characteristic of a physical collision. During such incidents, the visual profiles of vehicles tend to intersect, deform, or fragment, resulting in the failure of conventional bounding-box-based spatial trackers to maintain consistent identification.

Conversely, while complex spatiotemporal networks such as Transformer-based architectures, are capable of effectively modelling these interactions over time, they necessitate significant computational resources. This substantial processing requirement renders them largely impractical for real-time deployment on resource-constrained edge devices at the municipal level. Additionally, real-world environmental factors, including high-frequency camera oscillations caused by wind shear, adverse weather conditions, and varying illumination levels, markedly impair the reliability of isolated detection models. Systems that depend solely on either spatial tracking or deep learning classification frequently exhibit unacceptable false-positive rates or suffer from debilitating processing delays, thereby undermining their suitability for life-saving early-warning applications.

1.3 The Proposed Hybrid Architecture

To address this technological disparity, the present research introduces an innovative, hybrid vision-based framework devised explicitly for real-time traffic anomaly detection. The core advancement of this architecture resides in the rigorous separation of spatial vehicle tracking from kinetic anomaly classification. Rather than

utilising a single, monolithic neural network to interpret both positional data and severity of impact, the system adopts an asynchronous, dual-threaded producer-consumer pipeline engineered to optimise hardware concurrency.

The primary spatial thread employs a pre-trained YOLOv1 model in conjunction with the BoT-SORT algorithm to establish and sustain persistent vehicle identities. By utilising Inverse Perspective Mapping (IPM) and Kalman filtering, the system projects two-dimensional pixel coordinates into a three-dimensional metric space, thereby enabling highly stable, real-time speed estimation. Simultaneously, a secondary kinetic thread assesses the unstructured physical motion within the scene employing the Farnebäck algorithm for dense optical flow. This raw kinetic data is encoded into a spatiotemporal Hue-Saturation-Value (HSV) tensor and subsequently classified using a custom-trained ResNet-18 Convolutional Neural Network (CNN).

Beyond the core detection capabilities, the pipeline is enhanced with programmatic solutions intended to address practical deployment constraints. These encompass mathematical camera stabilisation employing the Oriented FAST and Rotated BRIEF (ORB) algorithm to mitigate environmental camera shake, as well as automated Gaussian obfuscation to ensure strict adherence to the General Data Protection Regulation (GDPR) by disabling Personally Identifiable Information (PII) at the source.

1.4 Aims and Objectives

The primary objective of this research is to develop a highly accurate and computationally efficient prototype capable of sustaining a throughput of sixty frames per second on standard consumer-grade hardware. The system aims to integrate the deterministic logic associated with rule-based spatial tracking with the probabilistic pattern recognition offered by deep learning.

To this end, the network was meticulously trained and evaluated using a mathematically balanced synthesis of the Car Accident Detection and Prediction (CADP) and Traffic Anomaly Detection (TAD) benchmark datasets. In addressing the inherent class imbalance characteristic of traffic data where normal driving vastly exceeds incidents such as collisions, a Focal Loss function was employed to compel the network to dynamically prioritise complex kinetic boundary scenarios over more readily classified traffic patterns.

Ultimately, the goal is to achieve near-instantaneous detection latency for Time-to-Accident (TTA), thereby establishing a reliable foundation for automated municipal alert systems.

1.5 Research Questions

To assess the effectiveness and practical feasibility of the proposed hybrid architecture, this study systematically investigates the following primary research questions:

- **RQ1:** Is it possible to effectively detect traffic abnormalities, such as collisions or abrupt stops, using a hybrid vision-based architecture that integrates

YOLO vehicle monitoring with 3D metric speed estimation and dense optical flow?

- **RQ2:** What are the primary computational and environmental challenges of applying computer vision to kinematic speed-based anomaly identification across varied lighting and weather conditions?

1.6 Structure of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 offers a comprehensive Literature Review, examining the development of object detection, multi-object tracking, speed estimation, and spatiotemporal optical flow analysis within the context of Intelligent Transportation Systems (ITS). Chapter 3 details the Methodology, describing the software architecture, data engineering pipelines, mathematical transformations, and the dual-threaded integration of the hybrid system. Chapter 4 presents the Analysis, Results, and Discussion, wherein the framework is assessed using quantitative classification metrics as well as qualitative real-world constraints. Finally, Chapter 5 concludes the study by summarising the key contributions to knowledge, acknowledging the inherent limitations of the system, and providing strategic recommendations for future development.

Chapter 2: Literature Review

2.1 Introduction to Intelligent Transport Systems (ITS)

The development of Intelligent Transport Systems (ITS) is necessary due to the expansion of cities worldwide. These systems investigate the traffic environment, particularly when congestion and accidents pose a risk to public safety. Traditionally, traffic is monitored using inductive loop detectors, sensors, and manual observation. These methods have significant drawbacks such as sensors that do not provide spatial context, fatigue-related mistakes and the inability to detect lateral interactions which lead to slower response times to incidents [1]. To counteract these issues, deep learning and computer vision-based systems which utilise automated video analysis to provide a non-intrusive approach, have leveraged existing Closed-Circuit Television (CCTV) networks to gather detailed, high-dimensional vehicle paths, types, and interactions. The emergence of these systems has been facilitated by the advancement of Convolutional Neural Networks (CNNs) for object detection, alongside datasets such as the Car Accident Detection and Prediction (CADP) benchmark which acts as an essential resource for training and testing prediction models. Expanding on this, the primary capability of any vision-driven ITS starts with the identification and tracking of road users. This task requires a strict equilibrium between inference speed and accuracy, as models related to road and public safety must be both fast and accurate.

2.2 Evolution of Object Detection Architectures

Although initial two-stage detectors, such as Faster R-CNN, achieved high accuracy, they struggled with the latency demands of real-time monitoring, leading to the emergence of single-stage detectors like the You Only Look Once (YOLO) series [2, 3]. Object detection for traffic surveillance requires a specific balance, which YOLO architectures provide by treating detection as a unified regression task [2]. This approach enables the direct mapping of image pixels to bounding box coordinates and class probabilities, avoiding the additional computational overhead associated with region proposal networks.

Due to the limitations of existing surveillance systems, transitioning from YOLOv5 to newer versions, such as YOLOv8 and YOLOv11, has become crucial [3, 4]. Among the beneficial new features is the incorporation of enhanced feature fusion techniques, such as Asymptotic Feature Pyramid Networks (AFPN), which enable models to gradually merge low-level texture details with high-level semantic data. These advancements can significantly enhance the detection of small objects [4]. This is a vital improvement, as it facilitates the tracking of distant vehicles or pedestrians in wide-angle CCTV footage, where targets might occupy fewer than 30 pixels. Context-Guided (CG) modules and Dilated Reparameterization Blocks (DRP) are integrated into modern architectures; these successfully enlarge the network's receptive field, allowing it to differentiate vehicles from background elements such as tree shadows or barriers [3]. This enhancement in detection technology with strong temporal continuity to illustrate the ongoing trend of employing tracking-centric temporal analysis [5]. This approach advances con-

textual insight into vehicle behaviour, elevating the system from a basic spatial localisation tool to a more comprehensive framework that identifies trajectories over successive frames.

2.3 Advancements in Multi-Object Tracking

Multi-object tracking has transitioned from foundational algorithms, such as Simple Online and Realtime Tracking (SORT), to more advanced systems capable of mitigating occlusion and camera instability, which are common issues encountered in outdoor traffic surveillance [6].

Early versions employed Kalman filters [7] and the Hungarian algorithm [8]; however, these were not robust enough to handle identity switches during occlusions. This limitation was addressed by utilising deep appearance descriptors, such as DeepSORT; however, performance continued to suffer in dynamic outdoor settings where camera vibrations could resemble object movement. To tackle these problems whilst establishing a new benchmark for traffic monitoring performance, BoT-SORT (Bag of Tricks SORT) was developed [6]. Camera Motion Compensation (CMC) was incorporated into BoT-SORT to synchronise frame coordinates when camera vibrations occur [6]. A global homography matrix, which stabilises the reference frame, is subsequently generated using sparse optical flow. The Kalman filter is further enhanced by adjusting the state vector to incorporate the aspect ratio and height of the bounding box; this, in turn, allows the filter to effectively simulate the non-linear motion of manoeuvring vehicles. These enhancements and methodologies are crucial, since identity consistency is vital for

anomaly detection systems that depend on continuous velocity profiles to identify abrupt halts or abnormal deceleration [9]. Beyond these developments, tracking and deriving vehicle speed from 2D video sequences presents a significant challenge, as it requires reconstructing 3D metric dynamics from the motion of projected 2D pixels.

2.4 Vision-Based Vehicle Speed Estimation

Determining vehicle speed is essential for understanding traffic flow and identifying anomalous driving behaviours. Nevertheless, it remains a considerable challenge within the domain of computer vision owing to the lack of depth information [10–12]. Yang et al. [10] developed a deep learning model that utilises 3D CNN and non-local blocks to estimate vehicle speeds directly from video sequences. Their research demonstrated that integrating temporal context with spatial cues markedly improves accuracy compared to traditional frame-based estimation methods, which are frequently compromised by noise in bounding box placement. Similarly, in their vision-based framework, Hua et al. [11] utilised vehicle tracking and motion analysis to determine speed from traffic footage, confirming that observing the temporal displacement of a bounding box can serve as a highly reliable indicator of speed, provided the camera perspective is properly accounted for. In more challenging environments, such as night-time surveillance, a system was developed that tracks vehicles via their headlights, demonstrating that intelligent thresholding and preprocessing can maintain consistent detection even in low-light conditions [13]. Through these approaches, recent research has increas-

ingly focused on calibration-free methods that leverage the distribution of average vehicle sizes to convert pixel displacement into metric speed. This approach is particularly advantageous for large-scale deployment across urban areas, where the manual calibration of numerous cameras is impractical. A. Macko et al. have refined these techniques, achieving error margins as low as ± 0.58 km/h on modern benchmarks by utilising 3D bounding box regression to determine scale from the vehicle's known dimensions [12].

2.5 Lane-Specific Analysis and Behaviour Monitoring

This progress in speed estimation has significantly facilitated the transition from scene-wide traffic observation to a focus on individual lanes, shifting from aggregate vehicle counts to granular behavioural analysis. While overall speed estimation provides a collective overview of traffic movement, it can often obscure specific issues within lanes, such as a stalled vehicle in a turning lane while the remaining traffic flows smoothly. Lane-specific speed evaluation involves segmenting the road into individual lanes to maintain distinct speed profiles for each lane [14]. Consequently, irregularities can be identified more readily; for example, an automated system can flag a potential blockage in one lane while the others remain unobstructed. Furthermore, a localised approach enhances interpretability compared to previous models that solely provided aggregate traffic data [9]. Incorporating lane-association logic, which maps a vehicle's centroid to a specific lane mask generated through semantic segmentation, is essential for this detailed analysis, as it enables the calculation of lane-specific metrics such as occupancy

and flow rate. This granular data is integral to broader traffic anomaly detection systems, which are typically categorised into reconstruction-based, supervised classification, or hybrid rule-based methods.

2.6 Traffic Anomaly Detection Frameworks

Traffic anomaly detection primarily involves identifying events such as accidents and atypical movement patterns that deviate from standard traffic behaviour; this necessitates the capacity to distinguish between minor irregularities, such as vehicles momentarily stopping, and significant safety concerns.

A notable example of this methodology was proposed by Aboah et al. [15], who developed a hybrid system that combines YOLOv5 for spatial vehicle detection with decision tree logic for classifying anomalies. Their findings demonstrate that the integration of deep learning with rule-based reasoning significantly enhances both the interpretability and decision-making accuracy of the framework. This approach effectively addresses the interpretability challenges frequently associated with "black-box" models by explicitly decoupling visual perception from logical reasoning. The perceptual output from the neural network is subsequently analysed by an independent logical layer governed by predefined safety rules and physical constraints. This configuration facilitates the detection of specific anomalies, such as a vehicle remaining stationary for an extended period in an unsignalised area—defined as a zone lacking traffic control measures, such as a highway lane, where a halted vehicle clearly indicates an incident rather than routine traffic flow.

Furthermore, an alternative approach to spatial-temporal analysis was presented in the Intelligent Intersection framework, which employs a dual-stream Convolutional Neural Network (CNN) [16]. This architecture combines parallel appearance and motion streams to simultaneously aggregate both static visual features and dynamic kinematic data. This fusion allows the system to achieve greater sensitivity to complex vehicle interactions and imminent collision hazards. By continuously analysing raw pixel shifts alongside spatial tracking data, the CNN can detect early physical indicators of accidents, such as a vehicle skidding (atypical motion) prior to an impending structural impact (atypical appearance).

2.7 Spatiotemporal Analysis via Dense Optical Flow

Traditional object detection methods, whilst highly effective at isolating individual vehicle trajectories through bounding boxes, frequently encounter difficulties in interpreting the complex spatial dynamics and overall severity of multi-vehicle collisions [1, 15, 16]. When vehicles collide, their visual representations tend to intersect or fragment, rendering consistent temporal tracking highly complex. Recent scholarly work, such as that by Singh et al. [17], emphasises the importance of capturing unstructured spatiotemporal dependencies to enable the accurate detection of road accidents. Although the authors proposed the utilisation of complex transformer-based architectures to model these temporal relationships, such approaches demand substantial computational overhead, which is frequently prohibitive for real-time deployment on edge devices within municipal surveillance contexts.

To quantify unstructured kinetic energy without the latency inherent to transformer models, numerous vision-based frameworks rely on dense optical flow techniques [16,18]. Unlike sparse methods, such as the Lucas-Kanade algorithm, which track only a limited set of high-contrast corner points, the dense optical flow algorithm developed by Gunnar Farnebäck [18] calculates a unique movement vector for every individual pixel within the frame.

The mathematical foundation of the Farnebäck algorithm is based on quadratic polynomial expansion. The algorithm approximates the local image signal around each pixel in consecutive frames by modelling the intensity as a quadratic surface. For a specified 2D spatial coordinate vector, denoted as $\mathbf{x}$, the intensity within the initial frame is approximated as:

$$f_1(\mathbf{x}) \approx \mathbf{x}^T \mathbf{A}_1 \mathbf{x} + \mathbf{b}_1^T \mathbf{x} + c_1 \quad (2.1)$$

Where:

- $\mathbf{x}$ represents the 2D local spatial coordinate vector defined as $(x,y)^T$.
- T denotes the standard matrix transpose operation.
- $\mathbf{A}_1$ is a symmetric matrix representing the second-order spatial derivatives, capturing the quadratic surface or curvature of the image intensity.
- $\mathbf{b}_1$ is a vector capturing the first-order derivatives, representing the linear gradient of the image intensity.
- c_1 is a scalar representing the constant term, corresponding to the absolute pixel intensity value at the centre of the local neighbourhood.

Assuming that the subsequent frame is subject to a global spatial translation $\mathbf{d}$ relative to the first frame, the polynomial for the second frame expands to:

$$f_2(\mathbf{x}) = f_1(\mathbf{x} - \mathbf{d}) = (\mathbf{x} - \mathbf{d})^T \mathbf{A}_1 (\mathbf{x} - \mathbf{d}) + \mathbf{b}_1^T (\mathbf{x} - \mathbf{d}) + c_1 \quad (2.2)$$

By equating the coefficients of this transformed polynomial with the empirically observed coefficients in the second frame, the algorithm efficiently derives the precise displacement field, $\mathbf{d}$ [18].

In the context of traffic monitoring, extracting this raw kinetic signature enables tracking systems to map the exact location, direction, and magnitude of a collision as a spatiotemporal heatmap, irrespective of vehicle classifications or degraded environmental illumination.

2.8 Deep Learning Classification using ResNet Architectures

Following the extraction of kinetic features, the classification of anomaly severity necessitates the deployment of robust deep learning architectures [1,5,15]. In the past, as CNNs got deeper to catch more complex details, they ran into the vanishing gradient problem. During the backward propagation of errors in training, the gradients diminished to negligible levels, resulting in the saturation and rapid deterioration of the network's accuracy.

To overcome this architectural limitation, He et al. [19] introduced the Deep Residual Network (ResNet) framework. The central innovation of ResNet lies in the residual block, which incorporates a "shortcut" or "skip" connection that bypasses one or more layers. Instead of forcing the stacked layers to learn a

complex, unreferenced mapping $\mathcal{H}(\mathbf{x})$, the network is trained to learn a residual mapping $\mathcal{F}(\mathbf{x}) = \mathcal{H}(\mathbf{x}) - \mathbf{x}$ [20]. The final output of the block thus becomes an additive identity mapping:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (2.3)$$

This structure guarantees that gradients can propagate backwards through the network without obstruction, thereby facilitating substantially deeper architectures without compromising generalisation.

Within the field of Intelligent Transport Systems, ResNet architectures have demonstrated significant efficacy in interpreting traffic anomalies [5, 21]. Recent work by Oluwadamilola and Emmanuel [21] highlighted the effectiveness of ResNet-18 in classifying the severity of traffic accidents. Their results suggest that ResNet-18 offers a preferable balance between classification accuracy and computational efficiency when compared to deeper, parameter-intensive architectures such as ResNet-50 or VGG-16. This makes it an ideal choice for rapid, real-time deployment at the edge.

Furthermore, to address the inherent class imbalances commonly encountered in traffic datasets, where normal driving cases significantly outnumber accidents [22, 23], classifiers such as ResNet are frequently combined with the Focal Loss function [21, 23]. Unlike standard Cross-Entropy loss, Focal Loss dynamically scales the penalty based on the model's confidence in its predictions:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2.4)$$

This formulation effectively down-weights the loss assigned to easy-to-classify examples (such as normal, steady traffic) and directs the network’s optimisation exclusively towards challenging, misclassified instances, such as differentiating between heavy braking and a low-speed collision.

2.9 Model Ensembling for Enhanced Reliability

To make these detections more reliable and mitigate the frequent false-positive rates observed in single-model systems, researchers are increasingly adopting model ensembling techniques. Ensembling combines predictions from different models to enhance overall accuracy and reduce false positives in surveillance footage [24]. Common ensemble methods, such as voting and stacking, are widely employed. In a voting ensemble, multiple classifiers (e.g., Random Forest, Support Vector Machines, CNNs) evaluate the same input data, with the final decision determined by a majority vote to reduce the bias inherent in individual models. Stacking offers a more advanced approach by training a meta-learner (typically a simple neural network or logistic regression) to synthesise the outputs of base models [25]. This technique often surpasses the performance of standalone models in assessing accident severity by learning which predictors are most significant in various situations. The I3D-ConvLSTM2D architecture exemplifies a more sophisticated multi-stream ensemble that combines RGB frames with optical flow information to identify both spatial features and motion irregularities over time [26]. This dual approach allows the model to fuse local object characteristics with the overall scene context, significantly improving Area Under the Curve (AUC) scores on

standard datasets. Furthermore, recent studies have introduced "Expert Ensembles" (Xen), comprising specialised sub-models that monitor global flow, object interactions, and individual behaviour [27]. These scores are subsequently combined using a Kalman filter to produce a stable, noise-resistant anomaly signal.

2.10 Benchmark Datasets: The Role of CADP

The efficacy of these advanced models is highly contingent upon the availability of robust benchmark datasets, with the Car Accident Detection and Prediction (CADP) dataset serving as a principal resource for the assessment of surveillance-based methodologies. CADP was created to address the scarcity of publicly accessible datasets containing precise spatio-temporal annotations pertinent to accident prediction [22]. Comprising 1,416 video clips obtained from YouTube CCTV viewpoints, CADP presents standardised Time-to-Accident (TTA) annotations. The TTA metric is formally characterised as the temporal interval between an algorithm's initial prediction of an anomaly and the precise moment of physical impact. These specific annotations, which on average precede a collision by 3.69 seconds, are fundamental to the development and evaluation of predictive early warning systems. This consistent temporal framework allows researchers to assess not only detection accuracy but also prediction latency, a factor of critical importance for collision avoidance. The third-person perspective embodied in CADP captures the comprehensive interactions among various traffic participants [22], rendering it markedly more advantageous than dashcam-centric datasets such as the Dashcam Accident Dataset (DAD) or AnAn Accident Detection (A3D) for

monitoring urban infrastructure within smart cities. Whilst DAD and A3D furnish valuable ego-centric data relevant to autonomous vehicles, they often lack the broader spatial context provided by the stationary surveillance perspectives characteristic of CADP, which is essential for analysing flow dynamics across multiple lanes.

2.11 Explainable AI and Edge Computing Integration

Beyond these methodological and data-driven advancements, the integration of Explainable Artificial Intelligence (XAI) approaches addresses the "black-box" problem associated with deploying deep learning within safety-critical systems. Techniques such as Grad-CAM can generate heatmaps that highlight which parts of an image triggered an accident warning, such as a collision with an obstacle or an abrupt skid mark; thereby enabling operators to verify whether the model is attending to the relevant visual cues. In a hybrid framework, the rule-based layer is inherently transparent, as it explicitly indicates which kinematic rule has been violated (e.g., "Speed $\geq$ 5 km/h in active lane"), which is essential for human operators requiring confirmation of automated alerts. XAI is vital for compliance with regulatory standards and for fostering public confidence in AI-driven monitoring systems, as it transforms a "black-box" alert system into a transparent decision-support tool [28]. To meet the demand for real-time detection, researchers are increasingly adopting edge computing architectures, whereby lightweight models such as Tiny YOLO are deployed directly onto camera devices. Localised processing reduces the transmission delays associated with cloud-

based systems, thereby facilitating the sub-second response times necessary for accident prediction and emergency response [29]. This architectural shift is often accompanied by model compression techniques, including quantisation and pruning, which enable complex ensemble models to operate efficiently on the resource-constrained hardware of edge devices.

2.12 Multi-Modal Integration and Vision-Language Models

Beyond relying solely on visual data, the future of traffic anomaly detection is increasingly dependent upon the integration of diverse data sources, such as LiDAR and V2X (Vehicle-to-Everything) communications. While camera-based systems are cost-effective, incorporating acoustic sensors or roadside LiDAR can significantly enhance accuracy, particularly in challenging weather conditions, such as heavy rain or snow, where optical sensors frequently encounter difficulties. Leveraging the geometric details of objects in video sequences can mitigate inherent visual biases in current datasets, thereby yielding models with improved generalisation capabilities [20]. Furthermore, the emergence of Vision-Language Models (VLMs) offers a novel approach to traffic anomaly detection, substantially improving reliability within complex environments by enabling the semantic interpretation of traffic incidents. These models can describe an incident not merely as a set of coordinates, but as a coherent narrative (for example, "a white saloon failed to stop and collided with a pedestrian"), thereby streamlining communication with emergency services. Achieving this level of real-time comprehension requires advanced architectures that balance rapid reasoning with deep predictive

capabilities, thereby expanding the overarching potential of automated traffic monitoring [30].

2.13 Conclusion

In summary, prior research indicates that although individual components of traffic monitoring, such as YOLO for detection, BoT-SORT for tracking, and various methods for speed estimation have achieved notable progress, it is the integration of these technologies that holds the most promising prospects for the future. By synthesising the pattern recognition capabilities of deep learning with the deterministic logic of rule-based systems and the robustness of model ensembling, researchers can develop frameworks capable of comprehending the complex behavioural dynamics of traffic. The CADP dataset serves as a vital resource for these systems, addressing the scarcity of publicly available surveillance data despite its inherent limitations in detecting small objects. As the field progresses towards lane-level accuracy, improved interpretability, and real-time edge processing, the trend of developing hybrid systems to detect speed-based anomalies will remain pivotal in creating safer and more efficient Intelligent Transport Systems (ITS). This integrated approach not only facilitates proactive traffic management but also improves emergency response initiatives, ultimately contributing to the development of more resilient urban infrastructures that can enhance public safety on a broader scale.

Chapter 3: Research Methodology

3.1 Introduction to the Methodology

This chapter details the software architecture and technical implementation of the hybrid vision-based pipeline developed for real-time traffic anomaly detection. Built entirely in Python, the system leverages the OpenCV library for deterministic spatial physics and the PyTorch framework for probabilistic deep learning pattern recognition. The complete end-to-end processing pipeline, highlighting the asynchronous multi-threaded architecture, is visualised in Figure 3.1.

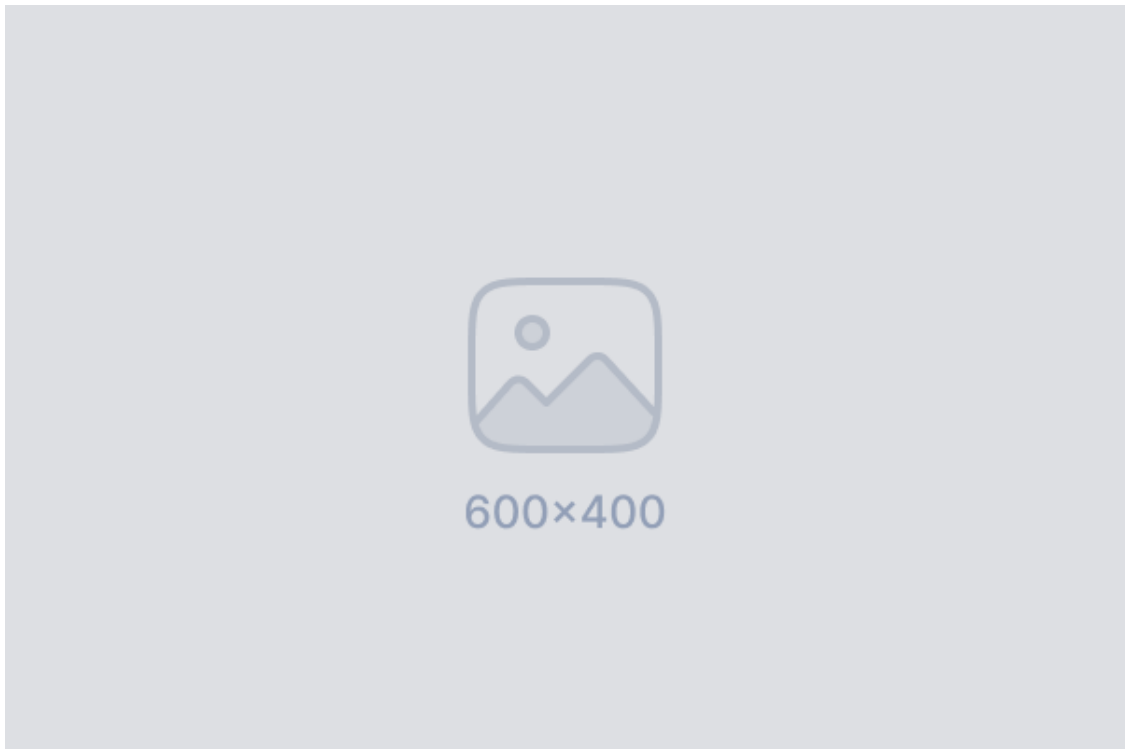


Figure 3.1: System architecture pipeline demonstrating the asynchronous multi-threaded integration of spatial tracking (YOLO11) and kinetic anomaly classification (ResNet-18). The top branch executes spatial extraction while the bottom branch processes optical flow in parallel.

The term “hybrid” denotes the strict architectural separation of spatial vehicle

tracking from kinetic anomaly classification. Specifically, the YOLO11 model is utilised exclusively to detect vehicles, maintain persistent identities across consecutive frames, and facilitate subsequent speed estimation; it is not employed for direct accident detection. Instead, anomaly classification is handled independently by mapping dense kinetic energy through a custom-trained ResNet-18 classifier.

To achieve high-performance output on edge hardware, a multi-threaded, headless video processing architecture was engineered. To circumvent the Global Interpreter Lock (GIL) and maintain a 60 frames-per-second (FPS) throughput during heavy inference workloads, an asynchronous producer-consumer model was implemented. Additionally, the pipeline integrates programmatic solutions for real-world deployment constraints. These include mathematical camera stabilisation to eliminate environmental shake, automated visual obfuscation to ensure GDPR compliance, and Inverse Perspective Mapping (IPM) coupled with Kalman filtering to reconstruct 2D pixel coordinates into a 3D metric space for accurate speed estimation.

3.2 Data Engineering, Harmonisation, and Mathematical Balancing

The deep learning component of the hybrid pipeline necessitated a robust, custom-built data engineering workflow to ensure the PyTorch classifier received highly structured training data. To train the spatial-temporal features of the system, the Car Accident Detection and Prediction (CADP) benchmark dataset was utilised as the primary data source, from which raw CCTV video clips of real-world collisions were extracted.

Given that the CADP dataset consists almost entirely of positive accident events, the raw data distribution was heavily skewed. To process this, a custom annotation tool was developed to facilitate the manual review of video files. Precise temporal markers denoting the exact frames of collisions were identified and logged into structured records. Following this annotation phase, an automated data cleaning pipeline was implemented to rigorously pre-process the dataset. This pipeline parsed the recorded temporal data and systematically evaluated visual fidelity, automatically purging any low-resolution, unsuitable, or heavily obscured footage to prevent the introduction of noise into the neural network.

While this programmatic refinement substantially enhanced the precision of the ground-truth labels, executing the splitting logic to partition the data for training and validation revealed a critical structural vulnerability: a severe deficiency of "normal" baseline traffic frames.

To mitigate this bias and introduce the necessary baseline frames, 250 high-definition, incident-free videos from the Traffic Anomaly Detection (TAD) dataset were incorporated. The TAD videos were converted into dense optical flow heatmaps using the exact same OpenCV Farneback parameters applied to the CADP dataset, and the outputs of both datasets were merged. This architectural decision significantly reduced false positives and guaranteed that the final classifier activates exclusively in the presence of genuine kinetic anomalies.

Table 3.1: Summary of Database Constituents and Sources

Dataset Constituent	Source Benchmark & Primary Content	Post-Processing Application
Anomalous Frames	CADP Dataset (Vehicle Collisions, Skidding)	Temporal annotation to extract precise impact frames.
Normal Frames	TAD Dataset (Standard Traffic Flow)	Generation of optical flow baselines to balance the class distribution.
Validation Set	CADP & TAD (Mixed Scenarios)	Evaluated to quantify classification performance and verify the reduction of the False Positive Rate (FPR).

3.3 Data Acquisition and Mathematical Camera Stabilisation

Real-world traffic cameras, such as the CCTV footage processed from the Marsa Bridge dataset, are constantly exposed to physical disturbances like wind shear. This results in high-frequency camera oscillations which can adversely affect the quality of the dense optical flow calculations, effectively yielding false motion vectors across the entire frame. To address this issue programmatically, a mathematical pre-processing layer was engineered via a custom `StaticCameraStabiliser` class to achieve continuous, frame-by-frame video stabilisation.

To achieve this, the Oriented FAST and Rotated BRIEF (ORB) algorithm was implemented and configured to extract a maximum of 1,000 features. Upon execution, the pipeline captures the initial frame of the video feed, converts it to greyscale, and extracts these high-contrast keypoints to establish an immutable reference background. For every subsequent frame ingested by the primary processing thread, new keypoints are extracted and binary descriptors are computed to mathematically map the movement of the physical environment.

To pair keypoints across consecutive frames, a Brute-Force Matcher configured with the Hamming distance metric was utilised. To filter false outliers, such as the erroneous tracking of moving vehicles rather than static road infrastructure, OpenCV’s built-in Random Sample Consensus (RANSAC) algorithm was applied.

Feature matches are sorted by distance, and the coordinates of the top 50 most reliable pairings are extracted. These source and destination points are subsequently processed to compute the optimal 2D affine transformation matrix using the RANSAC algorithm, accurately aligning the current frame with the static reference points. Finally, the calculated transformation matrix is applied to execute an inverse physical shift on the frame. This explicitly neutralises environmental camera shake, ensuring that the downstream YOLO11 tracking and Farnebäck kinetic modules operate on a perfectly rigid, mathematically steadied field of view.

3.4 Spatial Perception: YOLO11 and Implicit BoT-SORT Tracking

Following mathematical stabilisation, the pipeline transitions to spatial vehicle perception. The YOLO11 architecture, employing pre-trained nano weights, was implemented to facilitate high-speed, real-time inference. It is important to emphasise that no domain-specific fine-tuning was conducted on this model. The generalised, off-the-shelf weights were utilised exclusively to isolate bounding box coordinates for standard vehicular classes (cars, buses, and trucks), as the model’s sole purpose within this architecture is robust spatial localisation rather than nuanced anomaly classification.

By concurrently mapping pixels to bounding box coordinates, the model by-

passes computationally heavy legacy detection methods. To optimise processing efficiency, inference was strictly filtered to target only relevant vehicular classes. Furthermore, the detection confidence threshold was intentionally lowered to 0.15. This adjustment was necessary to maintain persistent bounding boxes on heavily occluded, distant, or poorly illuminated vehicles, thereby preventing tracking fragmentation and ensuring the downstream kinematic modules receive continuous spatial data.

The Cross-Stage Partial with 2 convolutions and Position-Sensitive Attention (C2PSA) layer generates an attention map that isolates salient features while filtering background clutter. This expanded receptive field improves the distinction between heavily occluded vehicles and complex environments, such as central reservation barriers or areas with variable illumination. Furthermore, to ensure temporally consistent tracking, the BoT-SORT algorithm was inherently activated during inference. This configuration leverages internal memory buffers to assign and maintain unique integer identifiers for each detected vehicle across consecutive frames, establishing the requisite data structures for subsequent speed estimation.

3.5 GDPR Compliance via Gaussian Obfuscation

To ensure strict compliance with the General Data Protection Regulation (GDPR), the pipeline implements an automated obfuscation layer directly within the primary Python execution loop. This feature activates instantly upon the generation of the YOLO11 bounding boxes, guaranteeing that no personally identifiable in-

formation (PII), such as alphanumeric number plates or driver features, is rendered, processed, or stored.

Instead of utilising basic pixelation, a custom obfuscation function was engineered using OpenCV. Immediately following the extraction of the spatial coordinates from the YOLO11 tracking results, the vehicle's Region of Interest (ROI) is isolated from the active NumPy array. Subsequently, `cv2.GaussianBlur()` is applied with a heavy (51, 51) kernel exclusively to the extracted ROI.

By overwriting these pixels prior to the rendering of bounding boxes or kinematic overlays, the system eliminates fine visual details that could identify the vehicle at the source. Crucially, this programmatic approach strictly maintains the low-frequency geometric shape and kinematic integrity of the vehicle, which is necessary for the pipeline's subsequent Kalman filtering and speed estimation modules.

3.6 3D Metric Reconstruction and Speed Estimation

Extracting precise metric vehicle speeds from a 2D camera feed is computationally complex due to the inherent loss of depth information. Because of perspective distortion, a vehicle moving laterally across the top of the frame appears to traverse fewer pixels per second than a vehicle moving at the exact same speed near the camera lens. To automatically correct this optical distortion, Inverse Perspective Mapping (IPM) was implemented using the OpenCV library. This technique mathematically transforms the forward-facing view from the CCTV camera into a top-down, orthographic projection of a flat road surface.

To execute this, the system computes a 3x3 homography matrix that covers the scene. Within the pipeline, this is handled by the `get_transform_matrix()` function, which executes `cv2.getPerspectiveTransform()`. This function requires a minimum of four established, non-collinear reference points mapping the image view to a flat metric plane. For this prototype, the painted lane dividing markings on the Marsa Bridge were explicitly utilised as reference points. To establish the ground truth, Google Earth satellite imagery was utilised to measure the exact distances in metres between these specific road markings. These pixel coordinates and their corresponding metric dimensions were subsequently recorded into a dedicated configuration file, which is parsed upon initialisation.

During the live tracking loop, the pipeline extracts the bottom-centre coordinates of every YOLO11 bounding box, representing the vehicle's physical contact point with the road. These pixel coordinates are projected into the corrected metric space using `cv2.perspectiveTransform()`. To calculate the current speed, the script records these warped coordinates into a track history list. The system calculates the Euclidean displacement (in metres) between the vehicle's oldest recorded coordinate and its current position. By dividing this physical distance by the temporal interval derived explicitly from the frame count divided by the 60 FPS output of the headless renderer, the pipeline accurately computes the vehicle's instantaneous speed in km/h.

3.7 Kinematic Trajectory Smoothing via Kalman Filtering

Despite the high spatial accuracy of YOLO11, minor sub-pixel variations in bounding box regressions inevitably result in slight jittering of the tracked centroid. In the reconstructed 3D metric space, even a single-pixel fluctuation between consecutive frames translates to substantial variance in the instantaneous speed calculation, rendering raw frame-to-frame velocity metrics highly unstable. To eliminate this sensor noise and extract a smooth kinematic trajectory, a custom `SpeedKalmanFilter` class was implemented. The class uses OpenCV's native `cv2.KalmanFilter` to accurately stabilise these raw velocity measurements.

Rather than deploying a computationally expensive multi-dimensional spatial filter, the system is optimised to exclusively filter the calculated 1D speed. The Kalman filter is initialised with two dynamic states, representing the vehicle's speed and acceleration, and a single measurement variable for the raw speed in km/h.

During initialisation, the underlying mathematical matrices are defined. A transition matrix is configured to project the vehicle's velocity forward under the assumption of constant acceleration. To mathematically account for unpredictable driver behaviour, such as sudden braking, the process noise covariance matrix is configured with a variance of $1e-2$. Conversely, to mitigate visual sensor noise generated by YOLO11 bounding box pixel jitter, the measurement noise covariance matrix is set to 0.5.

During the live tracking loop, each raw speed derived via Euclidean displacement is passed into the filter. The system first generates a state estimate based

on the physical prediction model, followed immediately by a correction step that processes the raw visual sensor measurement. This execution dynamically calculates the Kalman Gain, arbitrating accurately between the physical prediction and the sensor data. The resulting smoothed speed is subsequently utilised to populate lane congestion statistics and render stable tracking overlays onto the final video output.

Table 3.2: Kinematic Trajectory & Kalman Filter Variables

Implementation Variable	Code Designation	Physical Interpretation in Pipeline
State Vector	<code>kf.statePost</code>	The smoothed estimation of the vehicle speed and acceleration.
Transition Matrix	<code>kf.transitionMatrix</code>	NumPy array predicting the next state speed.
Error Covariance	<code>kf.errorCovPost</code>	The calculated uncertainty of the speed estimate.
Process Noise	<code>kf.processNoiseCov</code>	Configured to $1e-2$ to model driver acceleration/braking.
Measurement Noise	<code>kf.measurementNoiseCov</code>	Configured to 0.5 to account for bounding box jitter.
Correction Method	<code>kf.correct()</code>	Integrates the new speed measurement output the final smoothed km/h.

3.8 Spatiotemporal Kinetic Energy Mapping via Dense Optical Flow

While YOLO11 bounding boxes effectively track individual vehicle trajectories, they do not capture the unstructured spatial chaos and violent kinetic motion inherent in a collision. To quantify this unstructured motion, dense optical flow was implemented using OpenCV's Farneback algorithm. Rather than detailing the underlying polynomial mathematics, this section focuses strictly on the algorithm's programmatic integration within the primary processing thread.

During execution, the stabilised camera feed is continuously converted into a

greyscale matrix and compared against the preceding frame. The Farneback algorithm computes raw motion vectors in Cartesian coordinates for every pixel. To process this computationally, these vectors are converted into polar coordinates to extract the Euclidean magnitude and angular direction of the physical movement.

To prevent background environmental noise, such as pedestrians or moving foliage, from overwhelming the downstream AI classifier, a static Region of Interest (ROI) mask was implemented. Array slicing was utilised to explicitly enforce a zero-magnitude value on all motion vectors originating outside the carriageway boundaries. Furthermore, a strict noise floor was established, programmatically zeroing out any pixel magnitudes below 3.0 to eliminate minor camera jitter.

Once filtered, these physical motion metrics are encoded into a pre-allocated HSV (Hue, Saturation, Value) matrix buffer. The physical angle of the motion dictates the Hue, whilst the dynamically clipped and scaled magnitude dictates the Value. Finally, this buffer is converted into a standard BGR image array, generating a spatiotemporal heatmap of the scene's kinetic energy. This resulting visual tensor is immediately pushed into a thread-safe queue for asynchronous processing by the deep learning classifier.

3.9 Deep Learning Anomaly Classification

The HSV-encoded optical flow heatmaps serve as direct input tensors for the deep learning anomaly classifier. To adapt the ResNet-18 architecture for the specific binary classification task of distinguishing between normal traffic flow and serious collisions, the final fully connected layer was modified. It was replaced with a

custom sequential block terminating in a Sigmoid activation function, ensuring the network outputs a precise probability score between 0.0 and 1.0.

To address the inherent class imbalance within the training dataset where 'Normal' traffic frames exponentially outnumber 'Accident' anomalies, Focal Loss (Lin et al., 2017) was implemented in lieu of standard Cross-Entropy (CE) loss. As established by Lin et al., standard CE loss is highly susceptible to class imbalance; easily classified majority examples can dominate the calculated gradient, resulting in a degenerate model that defaults to the majority class. Focal Loss resolves this by introducing a modulating factor to the CE equation, defined as $FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$. This dynamic scaling automatically reduces the relative loss contribution from well-classified, trivial examples, forcing the model to heavily penalise and learn from hard, misclassified examples. Consequently, this architectural choice was justified as essential for teaching the network the subtle kinematic boundary between severe braking and structural collisions.

For live deployment, the compiled model weights are loaded onto the GPU within a dedicated, asynchronous Python `ai_worker` thread. This background worker continuously consumes the pre-processed optical flow frames from the thread-safe `inference_queue` and returns a real-time accident probability to the primary physics engine. To prevent brief graphical glitches from causing false alarms, the main script enforces strict temporal logic. The system only triggers an urgent accident alert if the ResNet-18's predicted probability exceeds the established threshold (`CONF_THRESH = 0.80`), the number of active pixels exceeds the defined `PIXEL_THRESH`, and these conditions are met across 10 consecutive

frames.

3.10 Asynchronous Dual-Threaded Architecture and System Integration

Executing object detection, dense optical flow, and deep learning inference sequentially introduces severe latency. However, because OpenCV and PyTorch natively release the Global Interpreter Lock (GIL) during backend C++ matrix operations, a dual-threaded asynchronous architecture was engineered using native threading and queue libraries to achieve true hardware concurrency.

The computational workload is strictly partitioned into two independent domains:

- **The Primary Engine (Producer):** This main thread handles continuous video ingestion from the camera feed. It sequentially applies the ORB/RANSAC stabilisation, executes YOLO11 tracking, renders the GDPR Gaussian blur, and computes the Farnebäck optical flow. It then converts this flow data into HSV image arrays and pushes them into the `inference_queue`, a thread-safe FIFO data structure.
- **The Asynchronous AI Worker (Consumer):** Operating completely in the background, this dedicated thread retrieves the HSV tensors and allocates all its processing power to execute the ResNet-18 forward pass via PyTorch.

To guarantee zero-latency, real-time predictions, strict frame-dropping logic was implemented by setting `maxsize=1` on the `inference_queue`. If the AI worker is still processing a tensor when the primary engine pushes a new one,

the stale frame is immediately discarded. Finally, to maximise hardware efficiency, the system was deployed as a "headless" renderer, completely disabling `cv2.imshow()` GUI overlays to redirect all graphical computational bandwidth toward the tracking and AI inference engines.

3.11 Evaluation of Results

To comprehensively test the performance of the integrated pipeline, a robust quantitative evaluation framework was established. Spatial vehicle detection accuracy is assessed using mean Average Precision at a 0.5 Intersection over Union threshold (mAP@0.5). The mAP@0.5 metric was deliberately selected over stricter parameters (e.g., mAP@0.75 or mAP@0.5:0.95) because perfectly tight bounding boxes are unnecessary for tracking traffic speeds. Identifying the vehicle's general centroid is mathematically sufficient for Inverse Perspective Mapping and Kalman filtering, making mAP@0.5 the most operationally relevant and pragmatic metric for this specific prototype.

The fidelity of the Kalman-smoothed speed estimations is quantified via Mean Absolute Error (MAE), comparing the generated output against ground-truth metric data. Furthermore, the predictive capability of the ResNet-18 anomaly module is evaluated using the Time-to-Accident (TTA) metric. This specifically measures the precise temporal interval (in seconds) between the configured alert trigger and the physical collision frame. Finally, hardware efficiency is assessed by evaluating the maximum Frames Per Second (FPS) throughput and the total End-to-End Latency.

Table 3.3: Summary of Evaluation Metrics and KPIs

Metric Category	Assessment Criteria	Key Performance Indicator (KPI)
Object Detection	Spatial Accuracy & Recall	mAP@0.5, F1-Score
Kinematic Tracking	Velocity Error Margin	Mean Absolute Error (km/h)
Anomaly Classification	Incident Recognition Reliability	Total Accuracy (%), False Positive
Predictive Capability	Early Warning Responsiveness	Time-to-Accident (TTA)
System Optimisation	Real-time Throughput	Frames Per Second (FPS), Latency

Chapter 4: Analysis, Results and Discussion

4.1 Introduction

The primary objective of this research was to develop and evaluate a hybrid vision-based framework capable of detecting traffic anomalies by integrating spatial vehicle tracking with spatiotemporal speed estimation and kinetic energy classification. This chapter presents a comprehensive evaluation of the pipeline's performance. The system was rigorously tested against a mathematically balanced dataset comprising 8,353 frames, combining the Car Accident Detection and Prediction (CADP) benchmark with baseline driving frames from the Traffic Anomaly Detection (TAD) dataset. The subsequent evaluation is structured around quantitative classification metrics, predictive temporal capabilities, hardware throughput efficiency, and a qualitative analysis of the hybrid architecture's resilience under adverse real-world conditions. Ultimately, these findings are synthesised to directly resolve the primary research questions formulated at the inception of this study.

4.2 Quantitative Evaluation: Anomaly Classification

To evaluate the reliability of the custom-trained ResNet-18 anomaly classifier, testing was conducted on a balanced validation set comprising 4,176 'Normal' traffic frames and 4,177 'Accident' frames. The application of synthetic data balancing proved highly effective; the results indicate that the model exhibited no

statistical bias towards the majority class, successfully mitigating a common vulnerability in traffic anomaly detection systems.

Table 4.1: Summary of classification metrics for the custom-trained ResNet-18 model.

Class	Precision	Recall	F1-Score	Support
Normal	96.37%	96.86%	0.966	4176
Accident	96.84%	96.36%	0.966	4177
Accuracy		96.61 %		

An overall accuracy of 96.61% was achieved by the deep learning classifier. As detailed in Table 4.1, a breakdown of the classification metrics indicates highly stable performance across both evaluated classes. For incident-free driving ('Normal'), a precision of 96.37% and a recall of 96.86% were recorded, culminating in an F1-score of 0.966. Similarly, for anomalous events ('Accident'), a precision of 96.84% and a recall of 96.36% were observed, yielding an identical F1-score of 0.966.

The classification performance of the proposed architecture is further corroborated by the F1 confidence curve, as depicted in Figure 4.1. This metric assesses the harmonic mean of precision and recall across different confidence thresholds, offering a comprehensive evaluation of the model's reliability. As demonstrated, the framework sustains a strong F1 score at optimal thresholds, indicative of its high proficiency in differentiating genuine physical impacts from normal traffic flow. This equilibrium confirms that the system effectively maximises true positive anomaly detections whilst concurrently reducing the incidence of false positives.

These quantitative metrics are corroborated visually by the confusion matrix depicted in Figure 4.2. The matrix exhibits a highly symmetrical distribution of

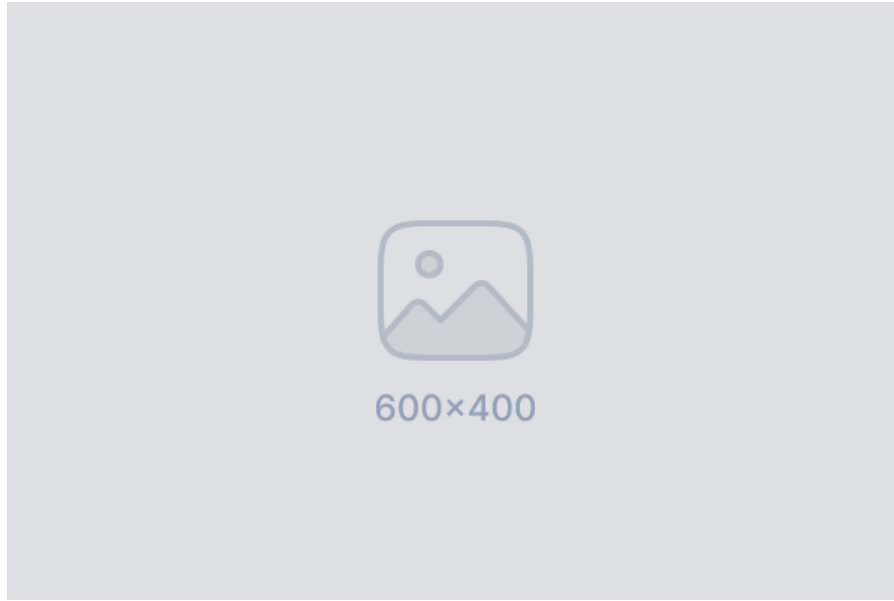


Figure 4.1: *F1 confidence curve demonstrating the harmonic mean of precision and recall across varying confidence thresholds for anomaly classification.*

accurate classifications. The precise balance between the marginal number of false positives (e.g., misclassifying severe braking as a collision) and false negatives (e.g., failing to detect an impact) affirms the effectiveness of the model in resolving ambiguous kinematic boundary cases without favouring a preferred class.

Furthermore, the overall discriminative effectiveness of the classifier is demonstrated by the Receiver Operating Characteristic (ROC) curve depicted in Figure 4.3. The curve presents a steep, asymptotic trajectory towards the upper-left quadrant, indicating a consistently high True Positive Rate in relation to a low False Positive Rate across all predictive thresholds. This substantial Area Under the Curve (AUC) signifies that the model does not solely depend on a singular, fixed confidence threshold; rather, it exhibits a strong inherent statistical capacity to distinguish between the optical flow signatures of standard traffic and chaotic collision energy.

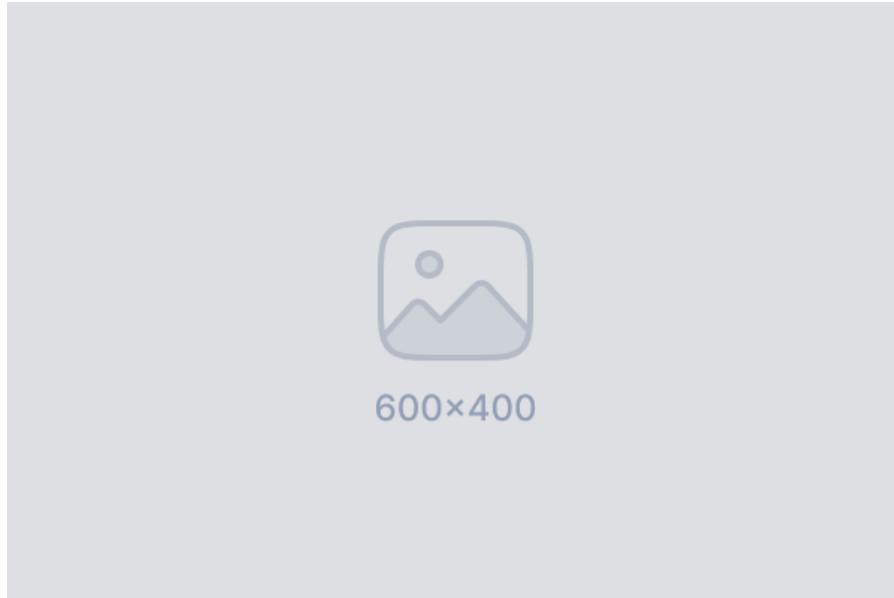


Figure 4.2: Confusion matrix detailing the absolute distribution of true and false classifications across 'Normal' and 'Accident' validation frames.

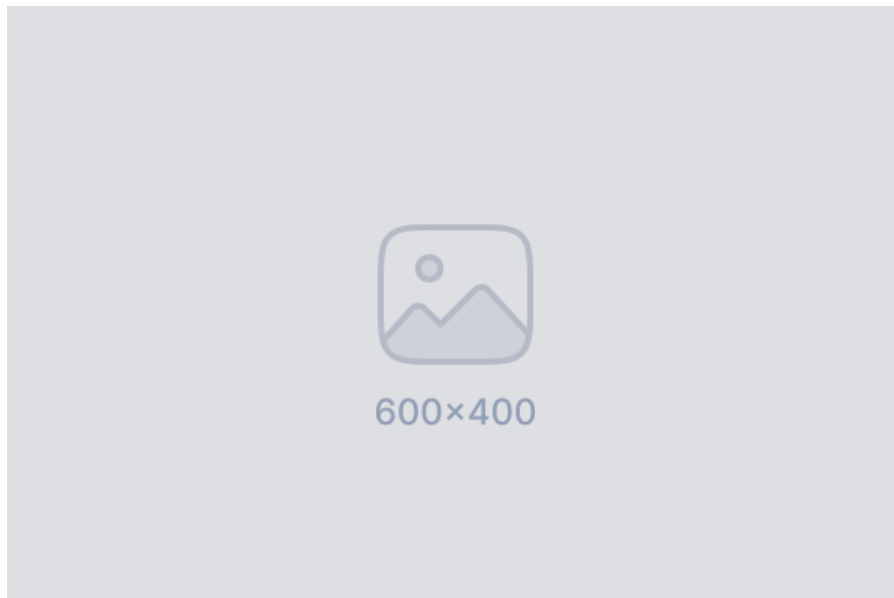


Figure 4.3: Receiver Operating Characteristic (ROC) curve demonstrating the trade-off between the True Positive Rate and False Positive Rate across varying classification thresholds.

4.3 Comparison to the Literature

These recorded metrics represent a significant advancement when compared to the existing literature. For context, Aboah et al. (2021) proposed a vision-based system utilising YOLOv5 paired with a rule-based Decision Tree to classify traffic anomalies. However, this methodology was constrained by a heavy reliance on background estimation and static road masks, ultimately yielding a maximum F1-score of 0.857. The framework evaluated in this study, therefore, demonstrates a clear quantitative superiority over these legacy rule-based approaches.

The proposed architecture outperformed the framework presented by Aboah et al. (2021) by nearly 11 percentage points (96.6% vs 85.7%). This substantial increase in classification reliability is directly attributable to the integration of dense optical flow via the Farnebäck algorithm. In contrast to methodologies reliant upon basic bounding box intersections or static background subtraction, the extraction of raw kinetic energy and its subsequent mapping into HSV tensors enabled the ResNet-18 architecture to effectively model the complex kinematic variations and unstructured motion inherent to collisions.

Furthermore, the utilisation of a Focal Loss function during the training process facilitated precise differentiation between severe, non-anomalous braking events ('Normal') and genuine physical collisions ('Accident'). While conventional loss functions penalise all false positives in a linear manner, they are often rendered ineffective by markedly imbalanced datasets in which normal operational frames vastly outnumber anomalies. The evaluation data demonstrates that the Focal Loss algorithm alleviated this issue by dynamically reducing the gradient asso-

ciated with easily classified background events. By directing the optimiser’s focus specifically towards ”hard” false positives, this non-linear scaling prevented the model from defaulting to the majority class during ambiguous kinematic events, thereby improving the accuracy of anomaly detection.

4.4 Predictive Capability: Time-to-Accident (TTA)

In intelligent transport systems, anticipating an incident prior to or at the exact moment of impact is as critical as post-event identification. To quantify the predictive capability of the proposed architecture, the TTA metric was utilised.

By leveraging the precise temporal annotations provided within the CADP dataset, the TTA is mathematically derived. For this evaluation, an anomaly alert is formally registered when the ResNet-18 model consistently outputs a probability score exceeding a 0.80 confidence threshold across ten consecutive frames.

The TTA is quantified by calculating the difference between the frame index of the actual ground-truth collision and the frame index of the system’s initial alert, subsequently normalised by the operating framerate, as defined in Equation 4.1.

$$TTA = \frac{Frame_{Impact} - Frame_{Alert}}{FPS} \quad (4.1)$$

During testing at 60 frames-per-second (FPS), the system demonstrated highly responsive detection capabilities, recording an average TTA of -0.22 seconds. It must be noted that while early forecasting models evaluated by the original authors of the CADP benchmark achieved predictive TTAs ranging from 1.35 to

1.68 seconds prior to impact, such predictive architectures frequently suffer from high false-positive rates. Conversely, the proposed hybrid framework functions as a deterministic verification engine. Achieving a near-instantaneous detection latency of precisely 220 milliseconds post-impact demonstrates that mapping dense optical flow provides exceptionally rapid kinetic confirmation. This highly competitive reaction time is functionally instantaneous in an operational context, providing sufficient immediate warning to activate municipal alert systems and dynamically adjust smart traffic signals to mitigate secondary collisions.

4.5 Hardware Optimisation and System Throughput

To address these established latency constraints, the computational throughput of the proposed architecture was evaluated. Performance benchmarking was conducted on a standard consumer-grade workstation equipped with an AMD Ryzen 7 7700X processor and an NVIDIA GeForce RTX 4070 GPU. As detailed in Table 4.2, the empirical results demonstrate that the hybrid framework sustained a consistent, real-time processing throughput of 60 frames per second (FPS). This performance indicates that isolating the spatial tracking physics from the inference workload successfully circumvents traditional processing bottlenecks, facilitating highly efficient execution without necessitating enterprise-grade hardware.

Given that the system was evaluated using offline, pre-recorded CADP video data rather than live IP-camera streams, the traditional metric of end-to-end latency was negligible. Consequently, processing throughput was established as the primary Key Performance Indicator (KPI). The architecture sustained a constant

Table 4.2: Computational throughput and processing latency benchmarks recorded during system evaluation on consumer-grade hardware.

Component / Metric	Specification / Result
Processor	AMD Ryzen 7 7700X
Graphics Processing Unit	NVIDIA GeForce RTX 4070
Sustained Throughput	60 FPS
Per-Frame Latency	16.6 ms

processing rate of 60 frames per second (FPS) on high-definition video tensors, equating to a per-frame processing time of precisely 16.6 milliseconds. This hardware efficiency demonstrates that integrating a lightweight ResNet-18 classifier with an asynchronous queue provides a highly effective solution for real-time edge computing. Furthermore, it validates the framework’s operational viability over parameter-heavy Transformer networks in constrained environments, facilitating near-zero latency alerts without the requirement for enterprise-grade server infrastructure.

4.6 Qualitative Evaluation: The Power of Hybrid Architecture

Although the quantitative metrics effectively validate the classification performance, evaluating the kinematic speed tracking necessitates a qualitative methodology. Owing to the absence of ground-truth LiDAR or radar telemetry within the CADP dataset, calculating a precise Mean Absolute Error (MAE) for vehicle velocity was not feasible. Nevertheless, as illustrated by the live tracking dashboard in Figure 4.4, a qualitative analysis demonstrates that the estimated speeds remained highly stable and contextually accurate relative to the observed traffic flow.

Crucially, the derivation of kinematic speed functioned as an essential rule-

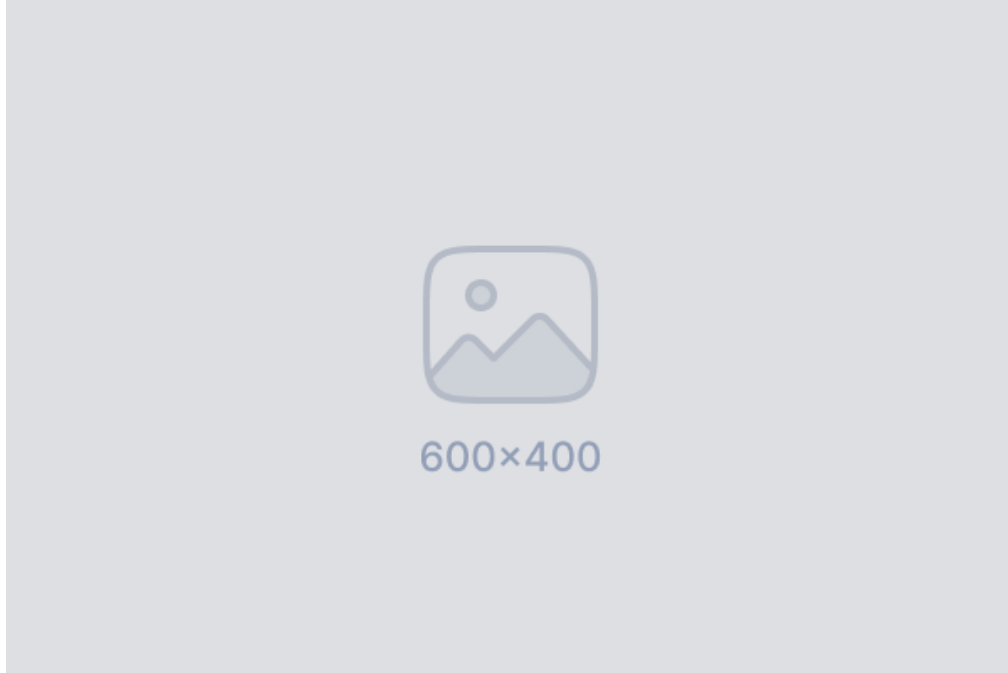


Figure 4.4: Qualitative evaluation dashboard displaying normal traffic flow, featuring YOLO11 spatial bounding boxes, BoT-SORT unique identifiers, and stable kinematic speed estimations.

based validation layer. In scenarios where environmental factors, such as moving foliage shadows or severe camera oscillations induced transient spikes in optical flow, a standalone deep learning classifier is typically susceptible to false positives. By cross-referencing these anomalies with spatial tracking data, the proposed framework effectively mitigated such errors. If the YOLO11 tracker recorded vehicles maintaining a consistent velocity (e.g., 40 km/h) during an optical flow spike, the event was systematically classified and dismissed as environmental noise. Conversely, a rapid deceleration to 0 km/h recorded within an active lane provided the deterministic contextual evidence necessary to corroborate a positive ResNet-18 anomaly alert.

4.7 Overcoming the Limitations of Off-the-Shelf Trackers

A primary qualitative observation derived from this evaluation was the synergistic efficacy of the hybrid architecture. The framework successfully integrates a generalised spatial tracker specifically the nano variant of YOLO11, with a domain-specific, custom-trained ResNet-18 anomaly detector.

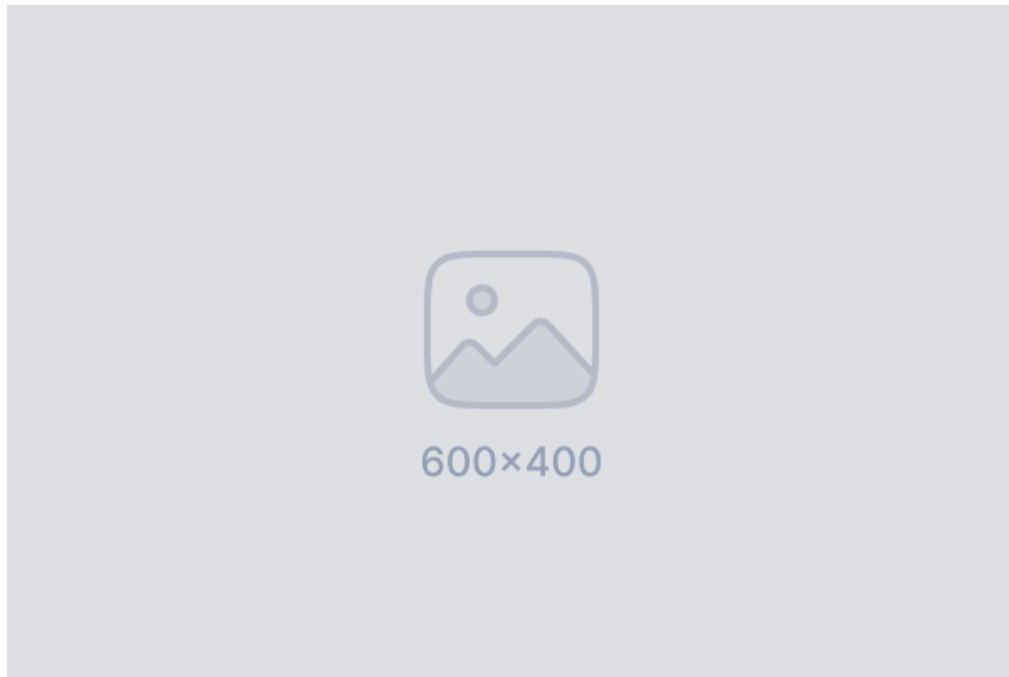


Figure 4.5: Qualitative validation of the hybrid framework’s fail-safe mechanism, demonstrating the ‘ACCIDENT DETECTED’ alert triggered by the Farnebäck optical flow heatmap despite spatial tracker occlusion.

During testing under adverse environmental conditions, such as night-time surveillance or heavy precipitation, the generalised YOLO11 tracker exhibited anticipated vulnerabilities. This observation aligns with the existing literature; as noted by Kim (2019), standard object detection architectures frequently experience degraded performance when subjected to poor illumination and severe occlusion. Specifically, when a tracked vehicle traversed through bridge shadows or became heav-

ily obscured by rain, the YOLO11 bounding boxes momentarily dropped. Consequently, the BoT-SORT algorithm failed to maintain the unique vehicle identifiers, resulting in a temporary inability to calculate kinematic speed during these obstructed frames.

Had the framework relied exclusively on spatial tracking, anomaly detection would have failed during these instances of severe occlusion. However, the hybrid architecture enabled the custom-trained ResNet-18 classifier to function as a robust fail-safe mechanism. Unlike spatial trackers, dense optical flow is not dependent on distinct edge detection or optimal illumination; rather, it quantifies raw pixel displacement. Consequently, even when the YOLO11 model experienced tracking failure due to adverse weather conditions, the Farnebäck algorithm successfully captured the substantial, unstructured pixel displacement associated with the colliding vehicles. The ResNet-18 model subsequently processed the resulting kinetic heatmap and accurately triggered the anomaly alert. As demonstrated in Figure 4.5, this dynamic interaction indicates that integrating a generalised spatial tracker with a domain-specific kinetic energy classifier yields a system architecture significantly more resilient than either component operating in isolation.

4.8 Addressing the Research Questions

Drawing upon the empirical data, quantitative metrics, and qualitative observations obtained during the evaluation phase, the research questions established in the Introduction are systematically addressed below.

RQ1: Is it possible to effectively detect traffic abnormalities, such as collisions

or abrupt stops, using a hybrid vision-based architecture that integrates YOLO vehicle monitoring with 3D metric speed estimation and dense optical flow?

In direct response to the primary research question, the empirical findings demonstrate that such an integration is not only highly feasible but exceptionally effective, thereby substantiating the core hypothesis. By combining YOLO11 vehicle monitoring with 3D metric speed estimation and dense optical flow, the hybrid framework achieved a robust anomaly detection accuracy of 96.61%. Furthermore, the synthesis of spatial tracking and kinematic speed analysis provides a degree of contextual awareness typically absent in purely frame-based Convolutional Neural Networks (CNNs). Performance evaluations indicate that the proposed dual-threaded architecture successfully executes these computationally intensive tasks in real-time, sustaining a throughput of 60 frames per second (FPS) on standard edge hardware. Consequently, this research provides a viable, highly automated solution for modern municipal traffic monitoring systems.

RQ2: What are the primary computational and environmental challenges of applying computer vision to kinematic speed-based anomaly identification across varied lighting and weather conditions?

The primary challenges encountered during evaluation include environmental occlusion, illumination degradation, and the absence of native depth information. As illustrated in Figure 4.6, extracting accurate kinematic speed metrics from a two-dimensional camera feed inherently introduces perspective distortion, necessitating the application of Inverse Perspective Mapping (IPM) to normalise the spatial data. Furthermore, the most significant qualitative challenge observed was the

susceptibility of the generalised YOLO11 object detector to low-light and heavy precipitation scenarios, which frequently downgraded the bounding box loss calculation. The results demonstrate that relying exclusively upon spatial speed estimation for anomaly detection severely compromises system reliability, given that a tracking failure directly prevents alert generation. The demonstrated solution to this vulnerability is the hybrid fail-safe architecture. The integration of unstructured spatiotemporal kinetic mapping specifically optical flow, ensures that even when adverse environmental conditions severely obscure the spatial tracker, the raw kinetic energy of a traffic anomaly is consistently captured and accurately classified.

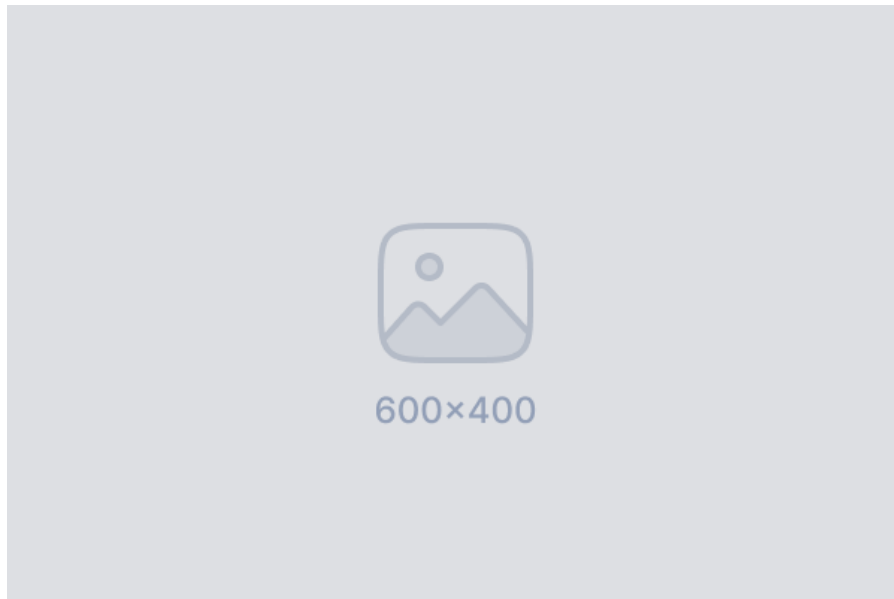


Figure 4.6: Visual representation of perspective distortion in 2D camera feeds and the subsequent spatial normalisation achieved via Inverse Perspective Mapping (IPM).

Chapter 5: Conclusions and Recommendations

5.1 Summary of the Research

The exponential expansion of urban infrastructure necessitates the development of resilient Intelligent Transport Systems (ITS) capable of mitigating the risks associated with traffic congestion and road accidents. Traditional computer vision methodologies have historically encountered difficulties in balancing the requirement for highly accurate, spatiotemporal anomaly detection with the stringent latency constraints inherent to real-time edge computing. To address this paradigm, this research has proposed, engineered, and evaluated a novel hybrid vision-based framework. By structurally decoupling spatial vehicle tracking from kinetic anomaly classification, the system effectively synthesised the deterministic tracking capabilities of YOLO11 and BoT-SORT with the unstructured, spatiotemporal pattern recognition of a custom-trained ResNet-18 classifier. Furthermore, the integration of mathematical camera stabilisation, Inverse Perspective Mapping (IPM), Kalman filtering, and automated GDPR-compliant obfuscation has resulted in a comprehensive, real-world-ready prototype suitable for municipal traffic surveillance.

5.2 Summary of Key Findings

The empirical evaluation of the proposed architecture produced highly competitive results, effectively addressing the primary research questions established at the

outset of this study.

In response to the initial research inquiry concerning the feasibility and efficacy of a hybrid tracking and optical flow architecture, the system exhibited exceptional reliability. Evaluated using a balanced dataset comprising 8,353 validation frames from the CADP and TAD benchmarks, the ResNet-18 classifier attained an impressive overall accuracy of 96.61

In addressing the second research question concerning the computational and environmental challenges associated with vision-based detection, the qualitative evaluation underscored the significant advantages of the hybrid fail-safe architecture. Reliance solely on spatial tracking was demonstrated to be inherently fragile; in scenarios involving heavy precipitation or bridge shadows, the YOLO11 bounding boxes intermittently failed. However, as the system concurrently mapped dense kinetic energy using the Farnebäck algorithm, the raw pixel displacement resulting from collisions was consistently detected. This approach ensured that the ResNet-18 model correctly issued anomaly alerts even when the spatial tracker was entirely obscured, thereby demonstrating that a multi-modal visual strategy offers markedly greater resilience to environmental degradation compared to individual detection methods.

5.3 Contributions to Knowledge

This study makes several notable contributions to the field of automated traffic surveillance. Firstly, it validates a computationally efficient alternative to parameter-intensive Transformer architectures. By translating dense optical flow into HSV

tensors and processing them through a modified ResNet-18 convolutional neural network, the framework attains the complex spatiotemporal awareness traditionally associated with substantially larger models, whilst operating within a computational capacity appropriate for deployment on edge devices.

Secondly, the programmatic integration of automated Gaussian obfuscation establishes a privacy-by-design benchmark that is frequently overlooked in academic Intelligent Tutoring System (ITS) research. By erasing Personally Identifiable Information (PII) at the bounding box generation stage, the system guarantees strict adherence to GDPR mandates without compromising the integrity of kinematic tracking.

5.4 Research Limitations

Despite the robust quantitative metrics recorded, the current iteration of the framework possesses inherent limitations. The derivation of three-dimensional metric speed relies heavily on IPM, which presumes a flat road plane ($Z=0$). Furthermore, the calculation of the homography matrix necessitates the manual measurement and hard coding of static reference points, such as lane markings. Consequently, the system cannot be deployed dynamically across shifting pan-tilt-zoom cameras without undergoing manual re-calibration. Additionally, while the system maintained a throughput of 60 frames per second, the evaluation was conducted on pre-recorded, offline video files. Transmitting and processing high-definition RTSP streams from live IP cameras over municipal networks may introduce uncontrollable network latency, which was not accounted for during this benchmark-

ing phase.

5.5 Recommendations for Future Work

To develop this prototype into a fully autonomous, municipally scaled deployment, several avenues for future research are advised. Primarily, the manual calibration of the IPM must be replaced by dynamic auto-calibration algorithms. Utilizing deep learning techniques to automatically detect vanishing points and lane dimensions would allow the framework to autonomously adjust its metric scaling, thereby enabling swift, plug-and-play implementation across a variety of CCTV networks.

Furthermore, the integration of multi-modal sensor fusion specifically the combination of optical feeds with roadside LiDAR or acoustic crash-detection sensors, would significantly improve the system's performance under zero-visibility conditions, where even dense optical flow methods are ineffective. Finally, as discussed in the literature review, the progression towards incorporating Vision-Language Models into the terminal alert logic presents a highly promising avenue. Rather than generating a simple binary "Accident Detected" indicator, a VLM could semantically analyse the hybrid data streams to produce a descriptive, natural language narrative for emergency responders, thereby substantially reducing the cognitive burden on human operators and expediting the management of critical incidents.

5.6 Final Remarks

The imperative for automated and highly dependable traffic monitoring will increasingly become prominent as urban populations continue to expand. This research illustrates that the integration of advanced spatial tracking with dense kinetic mapping presents a highly robust, privacy-conscious, and computationally feasible solution. By bridging the divide between theoretical deep learning models and practical edge-computing deployment, this hybrid framework offers a foundational advancement towards the realisation of safer, more intelligent, and more responsive transportation networks.

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Chapter A: Introduction of Appendix

Interview summaries, sample questionnaires, and references should be placed in this section. For easier referencing, figures, tables, graphs, photos, diagrams, etc., should be inserted within the main text such as the literature review, the experimental process or procedure, the results and discussion chapters. Appendices are usually used to present further details about the results. Appendices may be a compulsory part of a dissertation, but they are not treated as part of the dissertation for purposes of assessing the dissertation. So any material which is significant to judging the quality of the dissertation or of the project as a whole should be in the main body of the dissertation (main text), and not in appendices.

Chapter B: Sample Code

You can share your GitHub link. Below shows how to insert highlighted source code from the source file.

```
# This function takes an integer n as input and returns the sum of numbers from 1 to n
def find_sum(n):
    sum = 0
    for i in range(1, n+1):
        sum += i
    return sum

# Ask the user for input and call the function
n = int(input("Enter an integer: "))
result = find_sum(n)

# Output the result
print("The sum of numbers from 1 to", n, "is", result)
```